ZCAS University

BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING

CSS3211 SOFTWARE REQUIREMENTS ENGINEERING

FINAL EXAM

14:00 -17:00

Friday, 5th June 2026

TIME ALLOWED: THREE HOURS

INSTRUCTIONS:

1. Section A: ALL questions in this section are COMPULSORY and must be attempted
2. Section B: Answer any **THREE** questions only
3. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: All questions in this section are compulsory and must be answered

QUESTION 1

A university intends to develop an online course registration system.

a) Suggest one suitable software process model for the project.

b) Justify your answer with two reasons

c) Mention one process modelling tool that can be used during development. (4 marks)

d) Explain the meaning of a **software process**. Why is the software process important in software development? (4 marks)

e) Differentiate between the following terms:

a) Software Process and Software Product

b) Process Model and Development Methodology (4 marks)

f) State and explain any **four characteristics** of a good software process. (4 marks)

g) Describe the following software process models:

h) a) Waterfall Model

b) Spiral Model (4 marks)

i) Compare the **Waterfall Model** and the **Agile Model** using the following aspects:

a) Flexibility

b) Customer involvement

c) Testing approach

d) Delivery of software (4 marks)

j) Identify and explain any **four tools or techniques** used in process modeling. (4 marks)

k) What is a **flowchart**? Draw and explain a simple flowchart for a student login system.

(4 marks)

l) Explain the importance of **practical process modeling** in software engineering projects.

(4 marks)
(40 marks)

SECTION B: Attempt any THREE questions in this section.

Question 2

"SafeRide" is a new ride-sharing app with the following key requirements:

1. **User Authentication:** Riders must log in using email/phone and a 6-digit PIN.
2. **Fare Calculation:** The fare must adjust in real-time based on distance, traffic, and surge pricing (max 2x base rate).
3. **Safety Feature:** Riders can trigger an SOS alert, which shares their live location with emergency contacts.
4. **Performance:** The app must respond to SOS requests within 5 seconds, even under low network conditions (2G).*

a) Define **requirements validation** and explain how it differs from verification in this scenario. Why is traceability important when testing the *Safety Feature* requirement?

(4 marks)

b) Give **one example** of a *missing requirement* that testing might reveal in the fare calculation logic. (2 marks)

c) **Design test cases** for the *User Authentication* requirement (including inputs, expected outputs). **Identify ambiguities** in the *Fare Calculation* requirement and rewrite it for clarity. (6 marks)

d) Propose a **testing technique** (e.g., boundary value analysis) to validate the *Performance* requirement for SOS alerts. (4 marks)

e) A tester reports that the SOS feature fails under 2G networks. Explain how you would.

Trace the defect back to the original requirement.

Prioritize this bug versus a UI typo in the fare display (4 marks)

(20 marks)

Question 3

A commercial bank intends to develop a new Online Banking Management System that will support customer registration, account management, online transactions, loan applications, and mobile banking services.

Using your knowledge of software design, answer the following questions:

a) Define the term **software design** and explain the concept of **decomposition and modularity** in software design.
(6 marks)

b) Describe any two architectural styles that can be used in developing the banking system and discuss any three characteristics of a good software design.
(7 marks)

c) Explain any three issues that may arise during design creation and describe any two techniques used for improving software design quality. Explain the importance of design documentation in software engineering projects.
(7 marks)

(20 marks)

Question 4

A university plans to develop an E-Learning System that allows students to access learning materials, submit assignments, attend online classes, and communicate with lecturers.

Based on the scenario above, answer the following questions:

a) What is meant by **design evaluation and validation** in software engineering? Explain

the importance of **modularity** in the development of the E-Learning System? (6 marks)

b) Differentiate between the following architectural strategies:

c) i. Centralized architecture

ii. Distributed architecture

Discuss any **three characteristics of good software design** that should be considered when designing the E-Learning System. Explain three techniques for improving software design. (8 marks)

d) Explain three techniques for improving software design. Describe any two methods used in documenting software design. State any two reasons why design validation is important before implementation. (6 marks)

(20 marks)

Question 5

A software company has developed a Hospital Management System that will be used to manage patient records, appointments, billing, laboratory services, and pharmacy operations. Before deployment, the system must undergo comprehensive testing to ensure quality and reliability.

Using your knowledge of system testing, answer the following questions:

a) Explain the term system testing and state any two principles of system testing. Describe functional testing and explain its importance in the Hospital Management System. (7 marks)

b) Explain the purpose of **performance testing**. State any two performance factors that may be evaluated during testing. (3 marks)

c) Define the following terms as used in software engineering:

i. Reliability
ii. Availability
iii. Maintainability

What is **acceptance testing**? Explain any two types of acceptance testing.

(6 marks)

d) Describe the purpose of **installation testing** in software systems. Explain any two advantages of **automated system testing**. (4 marks)

(20 marks)

END OF EXAM